Bitcoin Ecosystem Layers and Monetary Stability

FROM BLOCKCHAIN THROUGH LIGHTNING NETWORK TO FREE BANKING

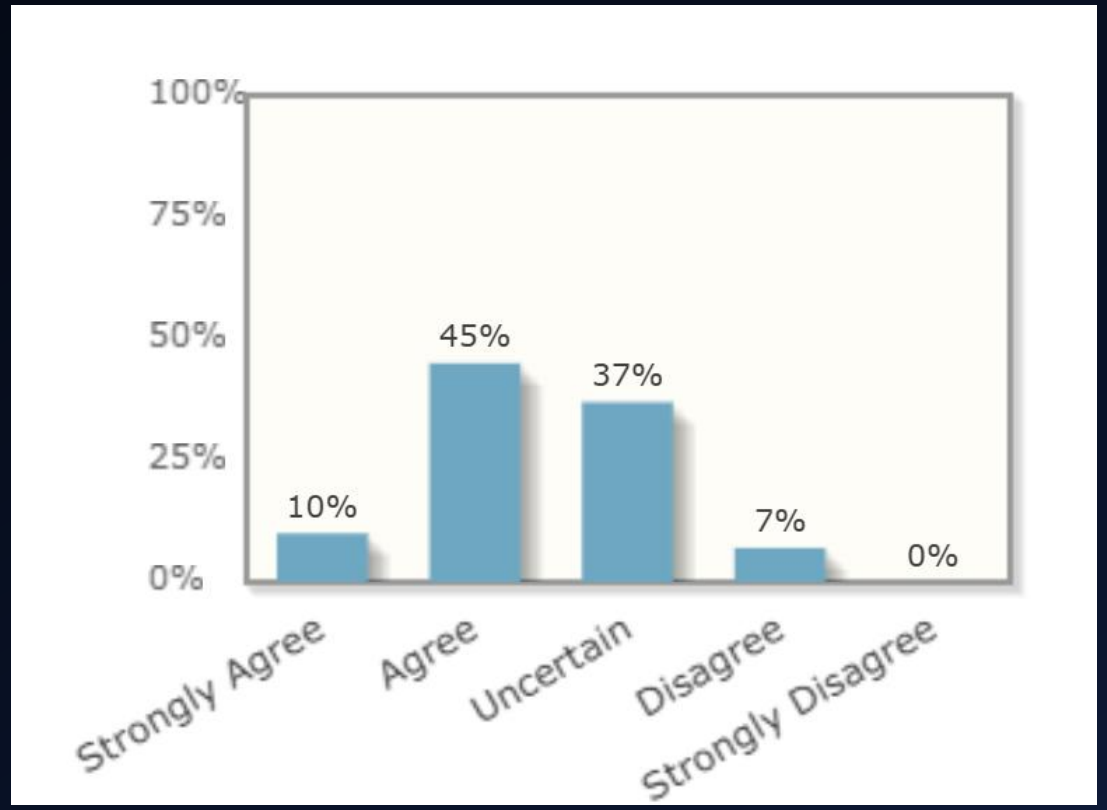
JIRI NOHEJL, GEORGE MASON UNIVERSITY

APEE 2025 MEETING, GUATEMALA CITY

Motivation

- “Given existing regulations, as **crypto assets grow in value** and become **more connected to the rest of the financial system**, the **fluctuations in their valuations** pose a serious **risk to financial stability** in advanced economies.”
- (Chicago Booth Kent A. Clark Center for Global Markets Survey, 2022)

- Confidence Weighted:



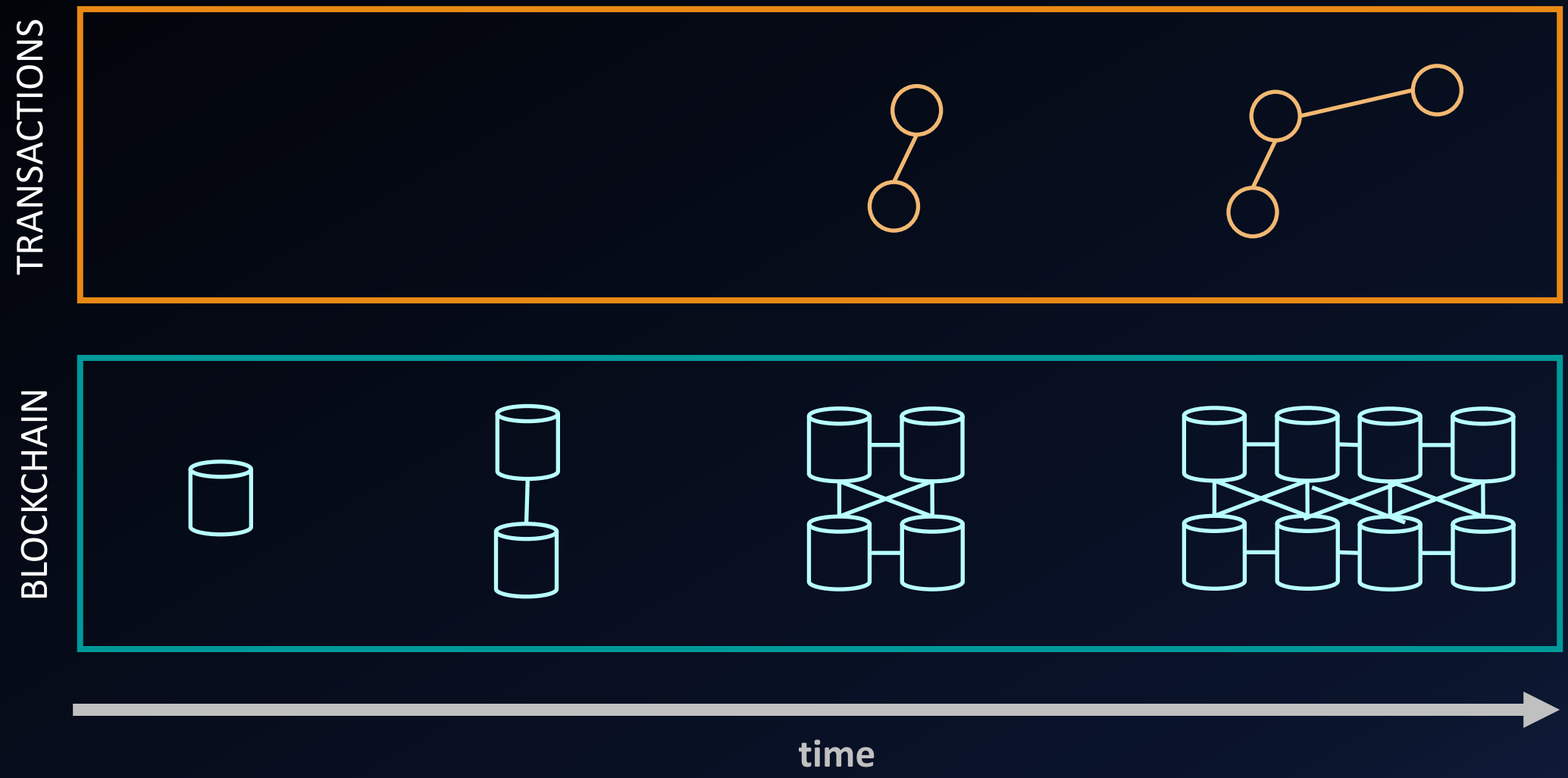
Motivation

- “[Bitcoin’s] obstacles in becoming a useful unit of account […] arises from its **extreme volatility**.” (Yermack, 2015)
- “[C]ryptocurrencies lack the stable institutional foundations needed to bolster the public’s trust […] making them **fragile and volatile**, as Bitcoin’s wild gyrations have amply demonstrated.” (Acemoglu, 2021)
- “[W]hat happens to [bitcoin] **value is basically irrelevant** to those of us not playing the crypto game.” (Krugman, 2021)
- “[I]t has become clear that Bitcoin is subject to **volatile prices and high transaction costs.**” (Prasad, 2021)

Problem: Time Inconsistent Bitcoin Launch

- How to guarantee "**honest nodes** collectively control more CPU power than any cooperating group of attacker nodes" (Nakamoto, 2009) on the newly establish network?
- If the Bitcoin nodes had a mechanism how to credibly commit to honesty the network could **align** its **supply growth with the growing transaction demand**.
- In the absence of such coordination mechanism what was good **solution for robust consensus** created **instability of bitcoin value** which decreased usability of bitcoins as "cash" for day-to-day transactions.

Problem: Time Inconsistent Bitcoin Launch



Purpose: Investigation into Layered Solutions

- Contrary to these prevalent views the wider adoption of Bitcoin might **under certain circumstances** dampen its future volatility.

> “I believe this will be the ultimate fate of Bitcoin, to be the "high-powered money" that serves as a reserve currency for **banks that issue their own digital cash**. Most Bitcoin transactions will occur between banks, to settle net transfers. Bitcoin transactions by private individuals will be as rare as... well, as Bitcoin based purchases are today.”
>
> Finney, Hal (2010)

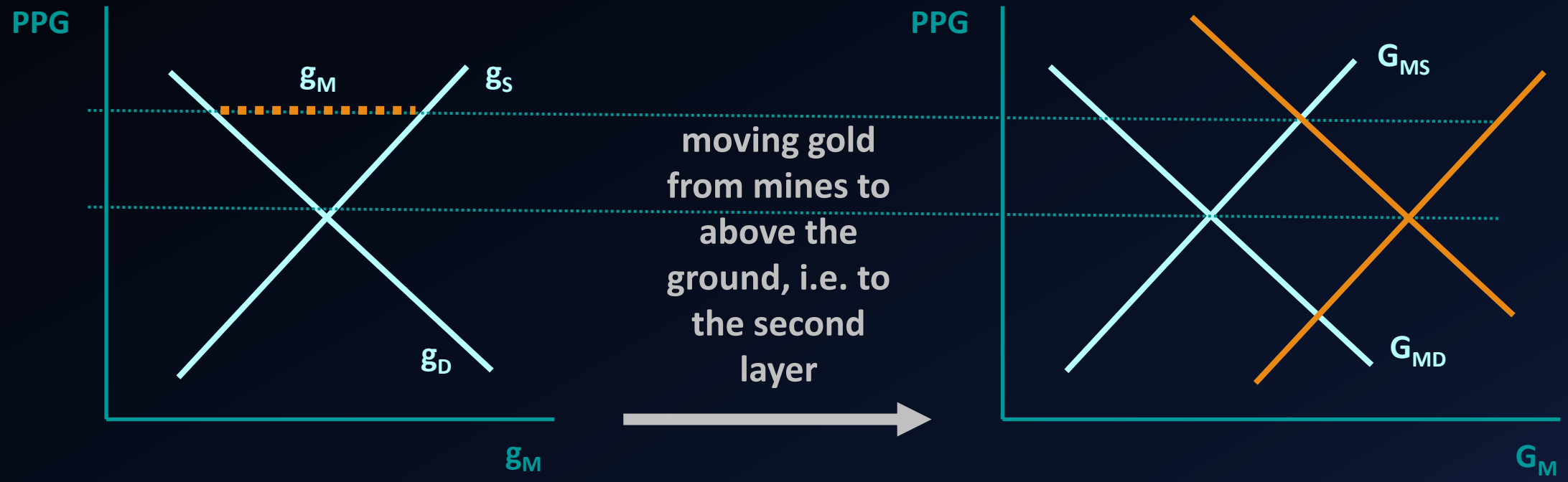
> “Hal Finney & I proposed this kind of thing, inspired by [George Selgin] and Lawrence White's historical research about free banking. We now have more **trust-minimized versions of this "layer 2" technology, e.g. Lightning**, which is fully & more securely collateralized than mere IOUs.”
>
> Szabó, Nick (2019)

- Why not **both**? And how that might work?

Gold Mining Stabilizes PPG (White, 2023)

"The increased monetary gold demand was ultimately satisfied almost entirely by an **enlarged stock of monetary gold** rather than by a permanent rise in the purchasing power of gold."

Lawrence H. White (2023) *Better Money: Gold, Fiat or Bitcoin.*

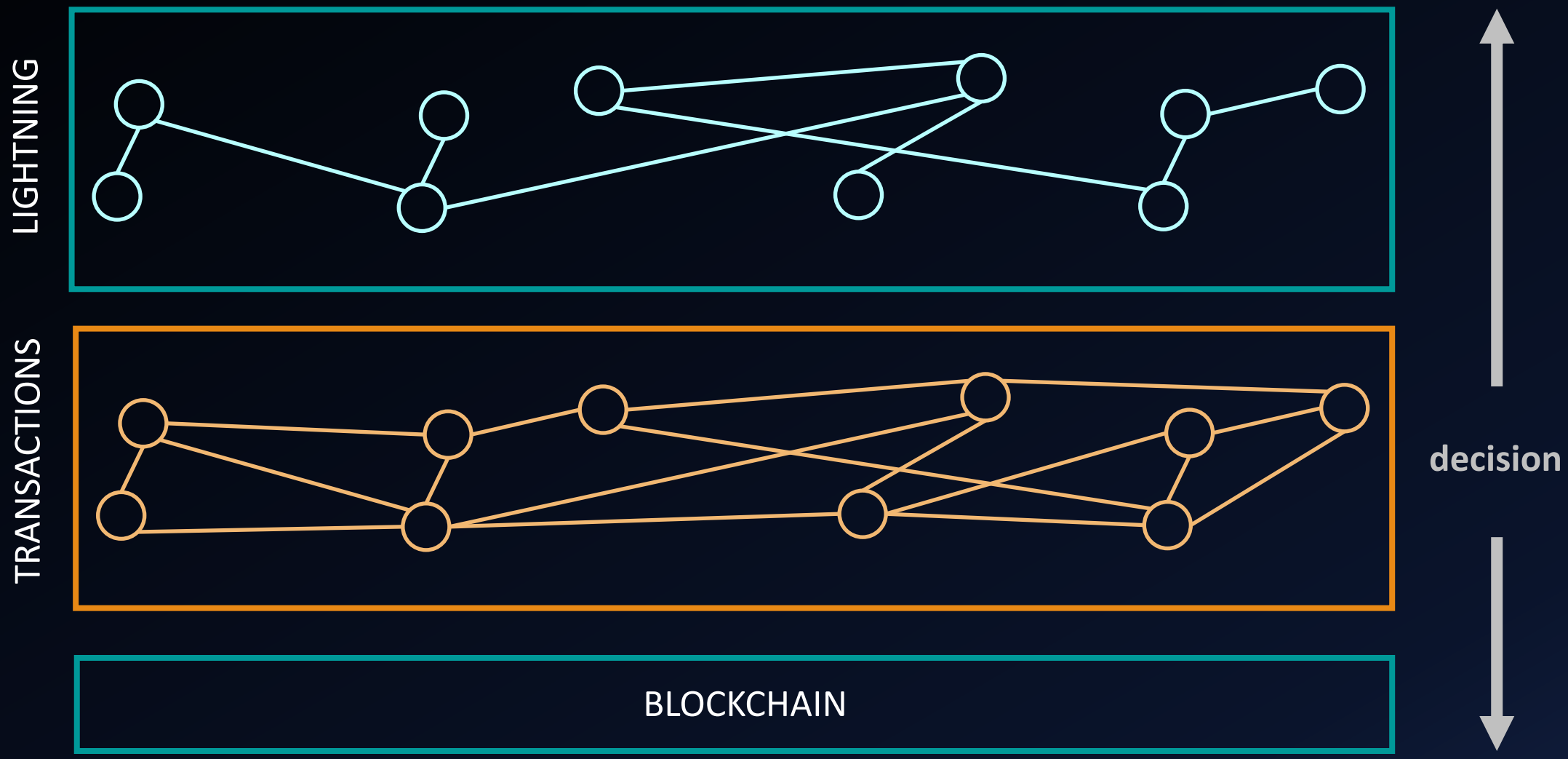


Blockchain - Lightning Network Dynamics

> "[The] usage of the Lightning Network is associated with reduced mempool congestion in Bitcoin and with lower fees [and] off-chain netting benefits of the Lightning Network can help Bitcoin to scale and function better as a means of payment."
>
> Divakaruni & Zimmerman (2022) *Federal Reserve Bank of Cleveland WP No. 22-19*

- Can decreased mempool congestion (high fees) through Lightning Network utilization lead to **increased stability?**
- **Alchian-Allen effect, RBF** and agents' decision making.
- Heterogenization of agents and separation of **demand to transact** and **demand to speculate**.

Blockchain - Lightning Network Dynamics



Free Banking on Gold Standard (White, 2023)

> "It is an attractive feature of **free banking** with fractional reserves that the quantity of bank-issued money expands when there is an **increase the demand to hold bank-issued money**, but does not expand otherwise (Selgin and White 1994, p. 1725). Free banking thus works **against short-run monetary disequilibrium** (excess supply or unsatisfied excess demand) and its business-cycle consequences."
>
> Lawrence H. White (2023) *Better Money: Gold, Fiat or Bitcoin.*

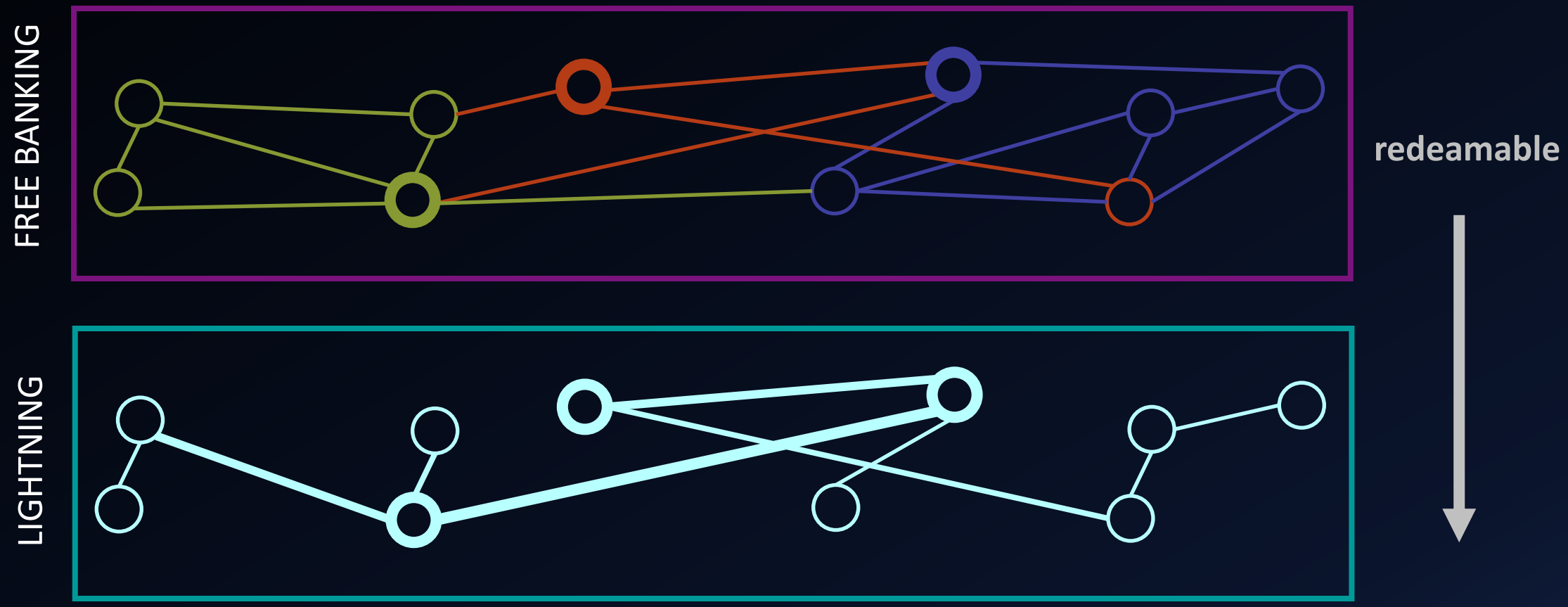
- Proto-banks issuing gold-redeemable "banknotes" evolved from **money-changers** and **goldsmiths**.
- These proto-banks were in their position given the excess supply of "underutilized" gold they held, and which could be used in **facilitating transactions more efficiently**. (Smith, 1776)

Lightning Network Structure and Free Banking

> "The analysis of [LN structure] has revealed a system whose **topology** has become increasingly characterized by star-like structures, **whose centers are constituted by 'hubs'** to which many nodes having a (much) small(er) degree, in turn, attach […] nodes have the incentive to become as central as possible, in order to **maximize [routing] fees** they may earn."
>
> Lin et al. (2022) *The Weighted Bitcoin Lightning Network.*

- Feature or a (decentralization) bug?
- John Holland's model of adaptive system (exploration x exploitation)
- Provides pathway towards third-layer where "centrality hubs" with their **high-capacity channels** can be understood as **proto-banks** emerging into institutions issuing **Bitcoin redeemable fiduciary media**.

Lightning Network Structure and Free Banking



Conclusion

- Growing value, acceptance and connectedness to financial system increases distinction between Bitcoin ecosystem layers with **emergent agents' heterogeneity**.
- Ultimately, there is **no conflict between trustless** robustness of Lightning Network **and trusting** free banking as such a multilevel solution proving ecosystem which can deal with problems which are not solvable on single layer.
- The dynamics between Blockchain, Lightning Network and Free Banking introduce **elasticity required to respond to money demand** fluctuations while keeping a **discipline of synthetic commodity monetary system**.

THANK YOU FOR YOUR ATTENTION